Methodology Paper

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A Human-in-the-Loop Strategy

A general framework and its application to the insurance industry

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Executive Summary

Artificial intelligence systems based on large language models, foundation models and agentic architectures are now sufficiently capable that organisations can plausibly delegate substantial portions of operational work to them. They are also sufficiently fallible, opaque and unevenly trustworthy that few responsible organisations will do so without human supervision. The discipline that governs how and where humans supervise, intervene in, learn from and remain accountable for AI behaviour is called human-in-the-loop, abbreviated HITL.

HITL is sometimes treated as a narrow operational pattern, equivalent to adding a confirmation button or a review queue. This document argues that HITL is better understood as an enterprise strategy. It determines which decisions an organisation is comfortable automating, which it requires a human to make, and which it expects the human and the machine to make together. It shapes the operating model, the assurance regime and the regulatory posture. In regulated industries it is now a precondition for deployment.

The document is in two parts. Part I sets out a general framework for designing a human-in-the-loop strategy. It defines the problem, classifies the principal interaction patterns, gives a graded decision rubric for choosing among them, and identifies the operating, technical, organisational and governance components that a credible strategy must include. Part II applies that framework to the insurance industry. It walks through the major insurance value-chain functions, identifies where HITL is essential, where it is helpful and where it is dispensable, and gives an indicative target operating model, governance structure and implementation roadmap.

> *The strategic question is not whether to keep humans in the loop. It is which humans, at which points, with what authority, and on the basis of what evidence.*

The headline conclusions are as follows. First, HITL design should follow the consequence of the decision, not the sophistication of the model. A high-capability model deciding something trivial needs less oversight than a moderate-capability model deciding something material. Second, HITL is not a single pattern; it is a portfolio. Mature organisations use human-in-command, human-on-the-loop, human-out-of-the-loop, and human-as-fallback patterns concurrently, matched to the decision at hand. Third, for insurance specifically, the appropriate distribution of human and machine effort differs sharply across underwriting, claims, fraud, pricing, customer service, distribution, and capital and reserving, and any uniform HITL policy will be both overcautious and underprotective at the same time. Fourth, regulators are converging on the requirement that AI-influenced decisions affecting consumers must be explainable, contestable and reviewable, and a credible HITL strategy is the simplest route to satisfying that requirement.

The document is intended for insurance executives, transformation leads, chief risk and chief data officers, and the AI engineering and policy teams that support them. It is methodological rather than prescriptive: the patterns and rubrics are reusable, the insurance application is illustrative of how those patterns translate when grounded in a specific industry.

PART I

A general framework for human-in-the-loop strategy

1. Defining the problem

1.1 What human-in-the-loop means

Human-in-the-loop refers to any deployment pattern in which a human participates in the operation of an AI system, whether by directing it, supervising it, correcting it, ratifying its outputs, or stepping in when it falters. The phrase has a long history in robotics and control engineering, but its contemporary meaning is shaped by the deployment of machine learning systems, and in particular by the deployment of large generative models whose outputs are neither fully deterministic nor fully verifiable in advance.

Three observations follow from that. First, HITL is not a synonym for manual review. Manual review is one specific instance of the broader pattern. Second, HITL is not necessarily slower than full automation; well-designed HITL processes are often faster than the manual baseline they replace, because the human handles only the small fraction of cases that benefit from human judgement. Third, HITL is not a transitional arrangement to be removed once the model improves. Some decisions require human authority for reasons that have nothing to do with model accuracy: legal mandate, fiduciary duty, ethical commitment, or the legitimacy of the decision in the eyes of the person affected.

1.2 Why HITL is now a strategic question

Three forces have moved HITL from operational design into strategic territory.

The first is capability. Foundation models can now produce drafts, summaries, classifications, recommendations and decisions across a wide range of business processes with sufficient quality that organisations can plausibly remove the human from many tasks. The cost of automation has fallen and the surface area of automatable work has expanded. The strategic question of where to keep humans is therefore live in a way it was not when machine learning was confined to narrow predictive tasks.

The second is failure mode. Generative and agentic systems fail in ways that are unfamiliar. They produce plausible but false outputs, comply with instructions that should be refused, behave inconsistently across superficially similar inputs, and on rare occasions take action sequences that no individual step would have indicated. The cost of being wrong about automation has therefore risen at the same time that the capability to automate has expanded.

The third is regulation. The European Union AI Act, sector regulators in financial services and insurance, and consumer-protection bodies in multiple jurisdictions have converged on the view that AI-influenced

decisions affecting individuals must be subject to meaningful human oversight, must be explainable, and must be contestable. The strategic posture an organisation takes on HITL is now visible to its regulators and, increasingly, to its customers.

1.3 What a HITL strategy is for

A human-in-the-loop strategy is the set of design choices an organisation makes about where, how and why humans participate in AI-augmented work. It serves four purposes.

1. It governs risk. By identifying the decisions for which human judgement is required and the form that judgement must take, the strategy converts a diffuse set of model risks into a defined operational control framework.
2. It allocates effort. By identifying the decisions for which human judgement is not required, the strategy releases scarce human attention for work where it adds genuine value.
3. It produces legitimacy. By making explicit how human authority is exercised and recorded, the strategy supports the legal, regulatory and reputational obligations that attach to decisions affecting customers, employees and third parties.
4. It produces learning. By treating human interventions as labelled feedback on model behaviour, the strategy creates a structured supply of training signal and a defensible audit trail of where the system is improving and where it is not.

> *A HITL strategy is not a safety net stretched under the AI. It is the operating model in which the AI sits.*

2. Patterns of human–machine interaction

HITL is a portfolio, not a single pattern. Four canonical patterns are useful for analysis. They are not mutually exclusive; mature deployments combine them, sometimes within a single process.

2.1 Human-in-command

The human makes the decision. The AI provides input. The human is the decision-maker in law, in process and in accountability. This pattern is appropriate where the decision has high consequence, where it requires the exercise of professional discretion, where contestability is important, or where the regulator requires a named human to be responsible.

Within human-in-command the AI may do substantial work: drafting documents, retrieving precedent, summarising large case files, suggesting options. The defining characteristic is that the human is not approving a machine decision; the human is making a decision with machine assistance.

2.2 Human-on-the-loop

The AI decides. The human supervises. The human does not approve each decision but monitors the system in aggregate, intervenes when patterns of error or drift are detected, and is empowered to stop the system or roll back its outputs. This pattern is appropriate where the volume of decisions is high, individual decisions are low-consequence, and an effective monitoring layer can detect aggregate problems faster than case-by-case review would.

The integrity of the pattern depends entirely on the quality of the monitoring layer. A nominal on-the-loop arrangement in which the supervising human has neither the time, the tools, nor the information to detect drift is no oversight at all.

2.3 Human-out-of-the-loop

The AI decides and acts without human involvement on any individual case. The human role is confined to system design, model selection, threshold setting, post-hoc auditing and incident response. This pattern is appropriate only where the decisions are reversible, low-consequence, well-understood, and where automation provides material value over the alternatives. It is also appropriate for time-critical decisions in which human latency would itself be the principal risk.

Out-of-the-loop is sometimes treated as the destination towards which all automation eventually moves. That framing is misleading. For many decisions, out-of-the-loop is not the destination at all; the destination is a robust on-the-loop arrangement, because the decisions do not warrant fully autonomous execution even when the model is capable of it.

2.4 Human-as-fallback

The AI decides until it encounters a case it cannot or should not handle, at which point the case is routed to a human. The defining characteristic is the existence of a triage or confidence mechanism that distinguishes the cases the AI handles from the cases it does not. This pattern is the most common in production deployments, because it allows the organisation to capture the efficiency of automation on the majority of cases while preserving human judgement on the minority that warrant it.

The pattern is only as good as its triage. A fallback mechanism that escalates too rarely produces silent errors; one that escalates too often produces the original manual process at higher cost. Triage design is therefore a first-order concern, not an implementation detail.

2.5 Combining the patterns

A single business process typically uses several patterns concurrently. A claims process may use human-out-of-the-loop straight-through processing for low-value digital claims, human-as-fallback for claims that exceed a threshold or fail an automated check, human-in-command for any claim involving suspected fraud or coverage dispute, and human-on-the-loop monitoring across all of the above to detect emergent bias or accuracy regression.

The strategic task is to assign each decision class to the right pattern, not to choose one pattern for the organisation.

The following table summarises when each pattern is appropriate.

Pattern	Description	Use when	Risk if misused
Human-in-command	Human decides; AI assists	Material decisions, regulated decisions, contestable decisions, discretion-heavy work	Wasted automation potential; bottlenecks; inconsistent decisions
Human-on-the-loop	AI decides; human supervises in aggregate	High-volume low-consequence decisions where aggregate monitoring is feasible	Theatre of oversight if monitoring is weak
Human-out-of-the-loop	AI decides and acts; humans only audit	Reversible, low-consequence,	Silent failure; unaddressed bias; loss of skill

Pattern	Description	Use when	Risk if misused
		well-understood, or time-critical decisions	
Human-as-fallback	AI decides until triage escalates	Mixed populations of cases with reliable confidence signals	Triage mis-calibration; escalation overload or under-load

3. Choosing the right pattern: a decision rubric

Assigning each decision class to the right pattern is the central design task of a HITL strategy. The choice should be driven by the decision itself, not by the model's capability. The following five dimensions provide a structured way to make that choice.

3.1 Five dimensions of decision character

The first dimension is consequence. How material is the outcome for the person, the firm or a third party? A decision whose worst plausible outcome is a small refund is qualitatively different from one whose worst plausible outcome is a denied insurance claim, a wrongful loan rejection, or a clinical misdiagnosis. Consequence determines how much oversight the decision warrants.

The second dimension is reversibility. Can the decision be undone if it turns out to be wrong, and on what timescale? A pricing error caught the next day is recoverable; a customer relationship lost to a wrongful policy cancellation may not be. Reversibility determines how much risk the organisation can tolerate before review.

The third dimension is contestability. Does the person affected by the decision have, or should they have, a right to challenge it? Decisions that are legally or ethically contestable require a named human in the decision chain who can answer for them, not because the AI is necessarily wrong but because contestability presupposes accountability.

The fourth dimension is frequency. How often is the decision made? A decision made a hundred times a day cannot reasonably take human-in-command form unless the cost of being wrong is very high. A decision made twice a year almost always can.

The fifth dimension is uncertainty. How confident is the model, and how reliable is its confidence signal? A decision the model is consistently confident about, and is empirically right about, is a stronger candidate for automation than one in which model confidence and model accuracy do not track each other.

These five dimensions can be combined into a simple rubric. The pattern that best fits a decision class is the one that takes its consequence, reversibility and contestability seriously while exploiting the available frequency and certainty.

Consequence	Reversibility	Contestability	Frequency	Indicated pattern
High	Low	High	Low	Human-in-command

Consequence	Reversibility	Contestability	Frequency	Indicated pattern
High	Low	High	High	Human-as-fallback with strict triage and named reviewer
Medium	Medium	Medium	High	Human-as-fallback or human-on-the-loop
Low	High	Low	High	Human-out-of-the-loop with audit
Low	High	Low	Low	Out-of-the-loop or no AI needed
Variable	Variable	High	Variable	Human-in-command as the default until evidenced otherwise

3.2 The capability fallacy

A common mistake in HITL design is to choose the pattern based on what the model can do rather than on what the decision requires. If the model is highly capable, the reasoning goes, then it can be trusted to act without oversight. This reasoning conflates two different questions: whether the model is technically competent to make the decision, and whether the organisation is willing for the model to be the author of the decision.

There are decisions for which the second question is settled regardless of the first. A regulator may require that a human underwriter sign off on a policy above a certain limit. A consumer may have a statutory right to a human review of an adverse decision. An organisation may have an ethical commitment that bereavement claims are never decided by an automated system. In each case the technical capability of the model is irrelevant to the design choice.

> *HITL design follows the decision, not the model. The question is not 'can the AI do this?' but 'do we want the AI to be the author of this?'*

3.3 Reversibility and the cost of being wrong

Reversibility deserves particular attention because it is often understated in HITL design. A decision is fully reversible if its consequences can be undone at modest cost and without lasting harm. It is partially reversible if some but not all consequences can be undone. It is irreversible if the consequences persist regardless of subsequent correction.

A wrong recommendation for a marketing email is fully reversible. A wrong premium quoted to a prospect is largely reversible if caught early, partially reversible if accepted by the customer, and may be irreversible if the customer has made decisions on the basis of the quote that they will not unwind. A wrongful claim denial that triggers a financial hardship event for the claimant is rarely fully reversible regardless of what the firm does afterwards.

The general principle is that the lower the reversibility, the higher the level of human oversight should be, regardless of how confident the model is in any particular case.

3.4 Contestability and the role of accountable humans

Contestability is the property that a decision can be challenged by the person it affects, that the challenge will be considered on its merits, and that the original decision can be revisited. It is a precondition for legitimacy in many contexts, particularly where decisions affect consumers, employees or other parties with limited power relative to the decision-maker.

Contestability requires a named human who has the authority and the information to revisit the decision. That human does not have to be the original decision-maker, but they must have access to the case file, the reasoning that led to the decision, and the authority to overturn it. In jurisdictions adopting Article 22 of the GDPR or comparable provisions, this is now a legal requirement for fully automated decisions with significant effect.

A HITL strategy that does not specify, for each automated decision class, who the accountable human is and what they have the authority to do, is incomplete.

4. Components of a credible strategy

A credible HITL strategy is more than a pattern map. It has six components, each of which has to be designed deliberately.

4.1 Decision inventory

The first component is an inventory of the decisions in scope. This is not a list of processes or systems; it is a list of the specific decisions that affect outcomes for customers, the firm or third parties. For each decision the inventory records its consequence, reversibility, contestability, frequency, the model or system that informs it, and the pattern under which it currently operates.

The inventory is the foundation. Without it, HITL design degenerates into pronouncements about the importance of human oversight that bear no defined relationship to any particular decision.

4.2 Triage and escalation design

Where the pattern is human-as-fallback, the triage mechanism is the determining feature. The triage may be confidence-based, rules-based, hybrid, or value-based, and the choice has significant operational consequences.

Confidence-based triage uses a calibrated model confidence score to decide whether the case is handled by the AI or escalated. It is operationally efficient but only as good as the calibration. Rules-based triage uses explicit thresholds on case characteristics, for example always escalating claims above a certain value, claims from new customers, or claims of certain types. It is transparent and auditable but coarse. Hybrid triage combines the two. Value-based triage uses an estimate of the expected cost of an error to drive escalation, escalating cases where the cost of being wrong is high regardless of model confidence.

The triage design should be reviewed regularly against ground-truth outcomes to ensure that the cases the system handles are the cases it actually handles well.

4.3 The interaction surface

The third component is the interface through which the human and the AI exchange information. This is the most underestimated element of HITL design. A well-designed surface lets the human see what the AI did, why, on what evidence and with what confidence, and lets them act on that information efficiently. A poorly designed surface produces rubber-stamping, in which the human approves the AI output because doing so is the path of least resistance.

A credible interaction surface presents the AI output, the salient evidence on which it was based, the model's confidence, the alternatives the model considered, and an unambiguous control for the human

to accept, modify or reject the output. It records the human's action and, where the human disagrees with the AI, the human's reasoning. It does not interrupt the human's workflow more than necessary, and it does not bury the AI's reasoning behind confirmation dialogs that train the human to dismiss them.

4.4 Monitoring and feedback

The fourth component is the monitoring layer that detects drift, bias, error and abuse, and the feedback loop that turns human interventions into training signal.

Monitoring should cover at least three dimensions. Accuracy: is the system getting the same decisions right at the same rate as it was at deployment? Bias: are error rates and outcomes distributed evenly across the populations the system serves, or is one group experiencing systematically worse outcomes? Drift: is the distribution of inputs to the system changing in ways that may invalidate its training? These metrics are computed on samples with known ground truth, including the cases that humans corrected, and reviewed on a defined cadence.

Feedback closes the loop between human action and model improvement. Every case in which a human overrules the AI is, in principle, a labelled example. Whether and how those examples are used to retrain or fine-tune the model is a design choice; that they should be captured systematically is not. A strategy that does not capture human overrides as data leaves substantial value on the table.

4.5 Roles, authority and accountability

The fifth component is the organisational structure that supports HITL operation. This includes the design of the front-line roles that work with the AI, the senior roles that own model performance and risk, and the governance bodies that set policy and adjudicate disputes.

At the front line, the roles should be designed for the work as it actually is, not as it was before the AI arrived. An underwriter working with an AI assistant is not the same job as an underwriter working without one; a claims handler reviewing AI-decided claims is doing a different job from one deciding them. Job descriptions, training, performance metrics and progression should reflect that. Failure to redesign roles is one of the most common reasons HITL implementations underperform.

Above the front line, there should be named owners for model performance, for HITL policy, and for incident response. In many organisations these sit with a chief data officer, a chief risk officer and a head of operations respectively. The principle is that no AI-influenced decision should be without an accountable human owner at each level of the organisation.

4.6 Governance and assurance

The sixth component is the assurance regime that demonstrates to internal stakeholders, regulators and customers that the strategy is operating as designed. This includes the policies that govern AI use, the standards that models must meet before deployment, the audit functions that verify ongoing compliance, and the incident management processes that handle failures.

Governance should be proportionate to risk. A low-risk customer-service summariser does not warrant the same scrutiny as a high-risk underwriting model, and treating them the same is itself a form of governance failure: it consumes assurance capacity that should be directed at higher-risk applications.

The six components are not independent. Decision inventory drives triage design, which drives interaction surfaces, which drive role design, which is monitored and governed. A weakness in any one component undermines the others.

5. Failure modes of HITL

Designing HITL well is harder than designing either fully manual or fully automated processes, because it has to manage the seam between the two. A small number of failure modes recur across deployments and are worth naming.

5.1 Oversight theatre

Oversight theatre is the failure mode in which the human appears to be in the loop but is not exercising meaningful judgement. The clearest signal is a high approval rate on AI-generated outputs combined with short review times. The human is rubber-stamping; the AI is effectively unsupervised.

Oversight theatre arises when the human is asked to approve outputs whose underlying reasoning they cannot evaluate, when they are asked to review more cases than they can plausibly examine, when their performance is measured on throughput rather than on the quality of their interventions, or when the consequences of rejecting an AI output fall on the human rather than the system.

The remedy is to design the human's role for what they can actually do. If the human cannot meaningfully evaluate the output, the AI should not be producing it without further safeguards; if the volume is too high, the triage should be tighter; if the incentives are wrong, the incentives should be changed.

5.2 Automation bias

Automation bias is the tendency of humans to defer to machine outputs even when their own judgement would have been correct. It is well-documented across decades of human-factors research and is not eliminated by warnings. It is exacerbated when the machine is confident, when the human is uncertain, when the human is under time pressure, and when disagreement with the machine carries a cost.

Mitigations include withholding the machine's recommendation until the human has formed an initial view, presenting evidence rather than recommendations where possible, designing performance metrics that reward considered disagreement, and ensuring that the cost of correctly disagreeing with the machine is no higher than the cost of incorrectly agreeing with it.

5.3 Skill atrophy

If the AI handles all routine cases and humans only see the difficult ones, humans may lose the practical skill base on which their judgement on the difficult cases depends. Over time this can hollow out the expertise the organisation relies on for the human-in-command parts of the portfolio.

Mitigations include deliberately rotating humans through the routine cases, maintaining training cases that exercise the full distribution of work, and ensuring that the AI's reasoning is sufficiently transparent that humans learn from working with it rather than being deskilled by it.

5.4 Triage failure

Triage failure occurs when the escalation mechanism routes the wrong cases to humans. There are two directions of failure. The first is under-escalation: cases that warrant human judgement are handled by the AI, and errors accumulate silently. The second is over-escalation: cases the AI could have handled correctly are routed to humans, and the efficiency gains of automation evaporate.

Both directions should be measured and bounded. The strategy should specify the acceptable rate of each kind of triage error and the cadence at which the triage is re-evaluated.

5.5 Diffuse accountability

Diffuse accountability occurs when something has gone wrong and no individual is the named owner. The AI made the recommendation; the human approved it; the policy permitted automation; the model was procured from a vendor. The strategy should remove this failure mode in advance by naming the accountable human for each decision class and each system, and by ensuring that those names are not so senior that the accountability is theoretical.

5.6 The shadow loop

A shadow loop is a HITL arrangement that exists on paper but is bypassed in practice. The reviewer approves without reviewing; the override is logged but not used in training; the monitoring dashboard is built but not watched. Shadow loops are often introduced when HITL is implemented in response to an external requirement rather than from internal conviction, and they fail at the moment the external scrutiny they were designed for arrives.

The remedy is to ensure that each HITL element is doing internal work for the organisation, not only documenting compliance with external mandate.

6. Implementing the strategy

6.1 A staged approach

HITL strategies are typically implemented in stages. A reasonable sequence is as follows.

5. Inventory: identify the AI-supported decisions in scope and characterise each one along the five dimensions of decision character.
6. Triage: for each decision class, choose the pattern, design the escalation mechanism, and specify the acceptable error rates.
7. Surface: design the interaction surface through which humans and the AI exchange information for each decision class.
8. Roles: redesign the affected roles, including job descriptions, training, performance metrics and progression.
9. Monitoring: instrument the systems to capture accuracy, bias, drift and the feedback loop from human interventions.
10. Governance: stand up the policy, assurance and incident-response capabilities, and connect them to the existing risk and compliance functions.
11. Operate and review: run the system, measure performance against the strategy, and revise on a defined cadence.

6.2 Metrics for HITL performance

A HITL strategy should be measured. The following metric families together provide a sufficient view of strategy performance.

- Decision quality: the rate at which AI-decided cases would have been decided the same way by a human, measured on a sampled basis with ground-truth labelling.
- Triage quality: the rate at which the cases routed to humans were the cases that genuinely required them, and the rate at which cases handled by the AI should in retrospect have been escalated.
- Human productivity: the throughput, cycle time and decision consistency of humans working with the AI, compared to the appropriate baseline.
- Disagreement rate: the rate at which humans overrule the AI, decomposed by reason, and tracked over time as a signal of model performance.
- Override outcomes: where humans overrode the AI, did their decisions prove correct? This is essential to distinguish principled overrides from automation aversion.

- Fairness: error rates and outcome distributions across the populations the system serves, monitored continuously and reviewed on a defined cadence.
- Contestability: the rate at which decisions are challenged by affected parties, the rate at which challenges are upheld, and the time to resolution.
- Incident rate: the rate, severity and time-to-resolution of incidents involving the AI system.

6.3 The build–buy–partner question

Most organisations will not build the underlying AI models. The strategic question is which components of the HITL stack to build in-house and which to procure. The models themselves and most monitoring infrastructure are typically procured. The decision inventory, triage design, interaction surfaces and role design are typically built in-house because they encode the organisation's specific risk appetite, regulatory posture and operational reality. Governance and assurance sit between the two, often built on procured platforms but configured to the organisation's specific policies.

6.4 Resourcing

Two resourcing patterns recur. The first is under-resourcing of the front-line roles: an assumption that introducing AI reduces the headcount required, leading to a process in which the remaining humans are too pressed to exercise the oversight the strategy formally requires. The second is under-resourcing of the assurance and monitoring functions, which are often treated as run-the-bank activities funded from existing risk and compliance budgets. The realistic position is that HITL implementation requires investment in both, particularly in the first eighteen to twenty-four months while the strategy is bedding in.

PART II

Applying the framework to the insurance industry

7. Why insurance

Insurance is a useful industry against which to test a HITL framework because its value chain spans the full spectrum of decision character. Some insurance decisions are high-frequency, low-consequence and reversible, and well-suited to automation under aggregate monitoring. Others are infrequent, irreversible and freighted with legal, ethical and regulatory significance, and require human authority regardless of model capability. The same firm typically operates both kinds of decision concurrently.

The industry has a long-standing relationship with quantitative models. Actuarial science, statistical underwriting and predictive modelling for claims have existed for decades, and the regulatory expectations attaching to model use, particularly under Solvency II in Europe, are mature. The introduction of generative and agentic AI does not therefore land on empty ground; it lands on an existing model-risk discipline, which is an advantage rather than a constraint.

Insurance is also a regulated consumer industry in most jurisdictions, with explicit fair-treatment obligations attaching to claims handling, pricing, distribution and complaints. In the United Kingdom the Financial Conduct Authority's Consumer Duty, in the European Union the AI Act and the Insurance Distribution Directive, and in the United States state-level insurance regulators each create requirements that bear directly on HITL design. The strategic question for an insurer is not whether to have a HITL strategy but whether the strategy it has will satisfy these regimes when tested.

7.1 The principal lines of business

The framework that follows is broadly applicable across personal lines, commercial lines, life and health. The specific configurations differ. Personal lines tend to be higher-volume, lower-value-per-policy, and benefit most from automation of routine decisions. Commercial lines tend to be more bespoke, with underwriting judgement preserved as human-in-command for material risks. Life and health introduce additional sensitivity around medical information, vulnerable customers and long-duration relationships, and tend to require more conservative HITL postures.

Where the discussion below refers to specific configurations, it is generally describing the personal lines case unless otherwise noted, on the basis that personal lines is where most of the automation opportunity and most of the regulatory exposure sit. The same logic applies in commercial lines and in life and health with the calibration adjusted.

7.2 Decisions versus processes

Insurance is often described in terms of processes: underwriting, claims, distribution, policy administration. These are convenient operational categories but they are not the right unit of analysis for HITL design. The right unit is the decision. An underwriting process contains many decisions, including risk acceptance, pricing, terms and conditions, reinsurance ceding, and renewal eligibility, each of which has its own consequence, reversibility and contestability profile. Treating them as a single process and applying a single HITL pattern across them is a category error.

The sections that follow are organised by process for readability but the analysis throughout is at the level of the constituent decisions.

8. The insurance value chain through a HITL lens

This section walks through the principal insurance functions and identifies, for each, the decisions in scope, their character, the indicated HITL pattern, and the practical considerations for designing the loop.

8.1 Underwriting

Underwriting is the decision to accept a risk, on what terms, and at what price. It is the canonical insurance decision and it is the one in which the AI-versus-human question is most contested.

8.1.1 The decisions

Underwriting comprises at least the following distinct decisions, each of which warrants its own analysis.

- Eligibility: does the prospect fall within the firm's appetite at all? Often expressible as rules and well-suited to automation.
- Risk classification: into which rating segment does the prospect fall? Driven by structured data, predictive models and, in some lines, by underwriting judgement on intangibles.
- Pricing: what premium reflects the risk? Driven by actuarial models, increasingly augmented by machine learning, and constrained by competition, regulation and conduct expectations.
- Terms and conditions: what exclusions, sub-limits and conditions apply? Typically discretionary in commercial lines, increasingly templated in personal lines.
- Reinsurance and capacity: should the risk be ceded, retained or shared? A capital-management decision rather than a per-policy decision in most personal lines but a per-risk decision in some commercial lines.
- Decline: a decision to refuse the risk, often contestable and increasingly regulated.

8.1.2 Indicated patterns

For personal lines high-volume products such as motor and home, eligibility, classification and pricing can reasonably operate out-of-the-loop within tight thresholds, with human-on-the-loop monitoring of aggregate outcomes and human-as-fallback for cases that fail automated checks or that fall in defined sensitive categories. Decline decisions, even at high volume, warrant human-as-fallback at minimum, because they are contestable and because misclassification at the decline boundary produces both regulatory and reputational risk.

For commercial lines, particularly above defined limits, underwriting is properly human-in-command. The AI's role is to surface evidence, structure the case, retrieve precedent and propose terms. The underwriter exercises judgement over the totality. Material risks should not be decided by automated systems even where the model is technically capable, because the legitimacy of the decision and the accountability of the firm require a named underwriter to own it.

For life and health underwriting, particular care is needed around protected characteristics, medical information and the use of inferred features. Models that use proxies for health status, family history or genetic information raise sharp regulatory and ethical questions, and HITL design should reflect that with conservative escalation defaults and explicit human review of any decline.

8.1.3 Specific considerations

Underwriting decisions are increasingly scrutinised for unfair discrimination, particularly where machine learning models use inputs that may serve as proxies for protected characteristics. The HITL strategy should include explicit fairness monitoring of underwriting outcomes by protected characteristic, conducted on a defined cadence with regulator-acceptable methodology, and an escalation mechanism if observed outcome disparities exceed defined thresholds.

Renewal underwriting deserves particular attention. The decision to non-renew, to apply a substantial premium increase, or to vary terms at renewal can be as consequential to a customer as an initial decline, and the regulatory expectation in many jurisdictions is converging on treating it the same way. The HITL pattern at renewal should not be materially weaker than at new business.

8.2 Claims

8.2.1 The decisions

Claims handling comprises a sequence of decisions of widely varying consequence and contestability.

- Notification triage: is the notification a claim, a query or a notification of circumstance? Mechanical and well-suited to automation.
- Coverage determination: is the loss covered under the policy? Often clear-cut but sometimes a matter of contractual interpretation.
- Liability assessment: in third-party lines, who is responsible? Driven by evidence, often contested by the other side.
- Quantum: what is the value of the loss? Driven by evidence and, in many cases, by professional estimation.
- Settlement: what payment is made, on what terms, and over what timescale?
- Reserving: what is the estimated ultimate cost of the claim, used in the firm's financial reporting?
- Recoveries: what amounts can be recovered from third parties, including subrogation and salvage?
- Suspected fraud: should the claim be referred for fraud investigation?

8.2.2 Indicated patterns

Straight-through processing of digital claims for low-value, well-evidenced losses is now the standard model for personal lines and is appropriate as a human-out-of-the-loop pattern within tightly defined

parameters. The parameters typically include claim value, customer profile, evidence quality, the presence of any fraud indicators, and the absence of any disputed coverage question.

Above the straight-through threshold, claims should move to human-as-fallback with the AI handling triage, evidence assembly and recommendation, and a claims handler making the substantive decision. The handler's interaction surface should present the AI's summary, the supporting evidence, the model's confidence, and any specific issues the model has flagged.

Bereavement claims, claims involving vulnerable customers, claims involving disputed coverage, and claims above defined value thresholds should be human-in-command. The AI's role is to assemble the file, draft correspondence and structure the decision; the handler decides.

Suspected-fraud referrals are a particular case. The decision that a claim warrants fraud investigation has significant consequence for the claimant, including delay to legitimate claims, and should be made by a human regardless of the AI's confidence. The AI's role is to surface the indicators, not to make the referral. The reverse failure, in which the AI clears claims that should have been flagged, is addressed by monitoring rather than by per-case human review.

8.2.3 Specific considerations

Claims is the area in which conduct regulators are most active, because it is the moment of truth in the insurance contract. A wrongful denial is one of the most damaging things a firm can do to a customer, and the firm's HITL strategy for claims will be examined accordingly. The strategy should be especially conservative on the denial decision: even where the AI has high confidence that a claim is not covered, a human should ratify any denial, and the reasons for the denial should be presented in terms the customer can understand and contest.

Reserving is a different kind of decision. It is not consumer-facing in the same way but it has material implications for the firm's financial reporting and capital adequacy. Reserving models are subject to actuarial standards and regulatory model-risk requirements, and the HITL pattern for reserving is typically human-in-command at the case-reserve level for material claims, with actuarial review of aggregate reserves on a defined cadence.

8.3 Fraud, financial crime and SIU

8.3.1 The decisions

Fraud detection has the unusual property that false positives and false negatives are both expensive. A missed fraud is a direct loss to the firm; a wrongful fraud accusation is a serious harm to a customer who, by hypothesis, is legitimate. HITL design has to balance the two.

- Score and rank: assign a fraud risk score to each claim or transaction. Continuous, model-driven, and operationally low-stakes per case.
- Investigate: refer the case to a fraud specialist for further enquiry. Material consequence for the customer; typically requires a human decision.
- Action: deny the claim, void the policy, or refer to authorities on the basis of fraud findings. High consequence and properly human-in-command.

8.3.2 Indicated patterns

Scoring and ranking can operate out-of-the-loop with human-on-the-loop monitoring for bias and accuracy. The decision to investigate should be human-as-fallback, with the AI surfacing indicators and the SIU analyst deciding whether to investigate. The decision to act on fraud findings is properly human-in-command and, in most jurisdictions, must be.

8.3.3 Specific considerations

Fraud models are an area in which fairness monitoring is particularly important. Historical fraud-detection systems have at times produced disparate impacts on protected characteristics, and the regulatory and reputational consequences of this are severe. The HITL strategy should include disparate-impact monitoring as a first-class element and not as an afterthought, and the threshold for human review should be tightened for any model whose disparate-impact metrics are not within defined tolerances.

8.4 Pricing, product and renewals

8.4.1 The decisions

Pricing decisions sit between underwriting and marketing. They include the technical price produced by the actuarial model, the commercial price quoted to the prospect, the renewal price applied at term, and the discretionary adjustments applied at any point. They are the area in which the regulatory focus on price walking, dual pricing and fair value has been most acute.

8.4.2 Indicated patterns

Technical pricing is properly human-on-the-loop. The model produces the price; actuaries and pricing teams monitor its performance, its calibration and its outcomes, and adjust the model. Commercial pricing involves the application of competitive and commercial overlays, which should be subject to clear policy, and any individual-level discretionary adjustments should be human-in-command with the AI providing the technical baseline.

Renewal pricing has been the subject of specific regulatory intervention in several jurisdictions. The HITL strategy should ensure that renewal pricing decisions are explicable, that the firm can demonstrate they are not the product of behavioural price discrimination against captive customers, and that vulnerable customer indicators are taken into account.

8.4.3 Specific considerations

Pricing is one of the areas in which the model itself is the regulator's primary concern, not only its output. Regulators in several jurisdictions now expect firms to be able to explain, at the level of an individual policy, why the price is what it is and how it would have differed under alternative inputs. HITL strategy in pricing therefore extends into explainability infrastructure: the ability to produce, for any quoted price, the principal drivers of the technical price and the principal adjustments applied to it. Without this capability the firm cannot operationalise contestability in pricing decisions even if its broader HITL strategy is otherwise sound.

8.5 Customer service and operations

8.5.1 The decisions

Customer service spans inbound contact handling, policy administration, mid-term adjustments, payment handling, complaints and retention. Many of these are now handled by conversational AI either as a first line or as a co-pilot to a human agent. The decisions vary widely in consequence.

8.5.2 Indicated patterns

Routine enquiries such as document requests, address changes, payment confirmations and policy summaries can operate out-of-the-loop within defined parameters and with effective monitoring of resolution quality. Mid-term adjustments that affect cover or price should be human-as-fallback at minimum, with the AI proposing the change and a human confirming. Complaints are a particular case: even routine-seeming complaints can mask vulnerability, and complaints handling should be human-in-command for the resolution decision regardless of the AI's role in triage and drafting.

Vulnerable customer identification is itself a HITL design problem. AI tools can be used to surface potential vulnerability from conversational signals, but the decision that a customer is vulnerable, and the resulting changes to how the firm treats them, should be human-in-command. The AI's role is to lower the threshold for human attention, not to make the determination.

8.5.3 Specific considerations

Conversational AI in customer service produces a particular HITL design challenge because the interaction is in real time and because the customer may not be aware they are speaking to an AI. The strategy should include a disclosure policy that is consistent with the firm's regulatory obligations, and a handoff design that allows the AI to escalate cleanly to a human when the conversation moves outside its competence. The handoff itself is a HITL design problem: a clumsy handoff that loses context, or one that takes too long, may be worse for the customer than a routed-to-human-first-time arrangement.

8.6 Distribution and intermediated business

Distribution introduces a third party into the HITL picture: the broker, agent or aggregator that sits between the firm and the customer. AI tools in distribution include broker-facing co-pilots, recommendation engines on direct channels, and the algorithmic systems behind comparison platforms.

The HITL question in distribution is partly about the firm's own systems and partly about the intermediary's. The firm is generally accountable for the outcomes its products produce for customers regardless of the channel, and its HITL strategy should therefore extend to its expectations of its intermediaries: what AI those intermediaries use, what their HITL arrangements are, and how the firm assures itself that the resulting customer outcomes are consistent with its own conduct expectations. This is a contractual and oversight question as well as a technology one.

Recommendation engines on direct channels should be human-on-the-loop at minimum, with the firm monitoring the quality of recommendations and the outcomes for customers who follow them. Personalised recommendations based on inferred customer characteristics warrant particular scrutiny, both because they may inadvertently treat similar customers differently and because they may produce sales of unsuitable products to vulnerable customers.

8.7 Capital, reserving and financial reporting

Capital modelling, reserving and financial reporting are areas in which the AI-versus-human question is constrained by professional standards and regulatory model-risk frameworks. Actuarial models are subject to Solvency II Pillar 2 requirements in Europe, to the equivalent state-level expectations in the United States, and to the IFRS 17 reporting framework globally. The introduction of machine learning into these areas is happening, but it is happening within an existing model-risk discipline that anticipates many of the questions HITL design asks.

The indicated patterns are predominantly human-in-command with on-the-loop monitoring of model performance. The chief actuary remains the named accountable human; the AI's role is to augment the actuarial process, not to replace the actuarial judgement. The HITL strategy in this area is therefore largely about integration with existing model risk management rather than introduction of a new discipline.

8.8 Summary: indicated HITL patterns across the value chain

The following table summarises the indicated HITL pattern for each principal decision class. It is illustrative; specific configurations will vary by firm, product line and jurisdiction.

Function	Decision class	Indicated pattern	Key consideration
Underwriting (personal)	Eligibility, classification, pricing	Out-of-loop / on-the-loop	Fairness monitoring; renewal parity
Underwriting (personal)	Decline	Human-as-fallback	Contestability; reasons in plain English
Underwriting (commercial)	Material risk acceptance	Human-in-command	Named underwriter accountability
Underwriting (life/health)	Decline; medical inference	Human-in-command	Protected characteristics; proxy risk
Claims	Straight-through low-value	Out-of-loop	Tight thresholds; clear evidence
Claims	Standard adjudication	Human-as-fallback	Interaction surface design
Claims	Denial, dispute, bereavement	Human-in-command	Conduct risk; plain-English reasons
Fraud / SIU	Scoring	Out-of-loop / on-the-loop	Disparate impact monitoring
Fraud / SIU	Investigation referral	Human-as-fallback	Cost of false positive to customer
Fraud / SIU	Action on findings	Human-in-command	Legal and regulatory exposure
Pricing	Technical model	On-the-loop	Model risk management

Function	Decision class	Indicated pattern	Key consideration
Pricing	Renewal pricing	On-the-loop with case escalation	Fair value; vulnerable customers
Customer service	Routine enquiries	Out-of-loop	Disclosure; clean handoff
Customer service	Complaints, vulnerability	Human-in-command	Identification ≠ determination
Distribution	Recommendation engines	On-the-loop	Suitability; intermediary oversight
Capital and reserving	Reserving, capital, reporting	Human-in-command	Chief actuary accountability

9. The target operating model

Applying the framework to the insurance value chain produces a set of decisions, patterns and assurance requirements. Operationalising them requires a target operating model that names the roles, the structures and the capabilities that hold the strategy together. This section sketches one. Specific firms will adapt it to their existing organisation.

9.1 Front-line roles

Front-line roles that work with AI should be designed around the work that humans are now actually doing, which is not the same work they were doing before. A claims handler in a HITL operating model is not handling fewer cases of the old kind; they are handling a different population of cases, with different evidence in front of them, different tooling, and a different relationship to the AI's output.

9.1.1 The AI-augmented underwriter

The role spans risk evaluation, terms-setting, pricing oversight and broker relationships. The AI assistant retrieves precedent, summarises submissions, drafts terms and flags anomalies. The underwriter exercises judgement on appetite, terms and price for material risks, and reviews escalations from the automated portfolio. The metric set rewards considered decision quality, not throughput; the training programme includes deliberate exposure to the full distribution of cases, not only the ones the AI escalates.

9.1.2 The AI-augmented claims handler

The role focuses on the cases that warrant human attention: complex liability, disputed coverage, vulnerable customers, suspected fraud and adverse decisions. The AI handles triage, evidence assembly and routine adjudication. The handler's interaction surface presents the AI's draft decision, its supporting evidence, its confidence, and any specific flags. The role's metrics include decision quality, customer outcome and override outcomes, with throughput as a hygiene measure rather than a primary metric.

9.1.3 The vulnerability and complaints specialist

This is a role that often grows in HITL operating models. The AI surfaces potential vulnerability and potential complaints from conversational data; the specialist makes the determination and decides on the resulting handling. The role is, in effect, the operationalisation of conduct risk in the front line, and is one of the clearest illustrations of the principle that the AI lowers the threshold for human attention while the human retains the decision.

9.1.4 The pricing and underwriting analyst

The technical pricing and underwriting analytics function expands in a HITL operating model because it owns the on-the-loop monitoring of the automated portfolio. The function monitors accuracy, fairness, drift and disparate impact across the automated decisions, investigates anomalies, and is the first line of escalation when the monitoring layer detects a problem. It is also the function that owns the feedback loop from human overrides into model retraining.

9.2 Second-line and risk roles

Second-line functions, including risk, compliance and model risk management, take on additional responsibilities in a HITL operating model.

9.2.1 Model risk management

Model risk management is mature in insurance because of Solvency II and equivalent regimes. The HITL strategy expands its scope from actuarial and capital models to all AI systems that influence customer outcomes. The function owns model inventory, model validation, ongoing model performance review and the model-risk policy framework. The HITL strategy depends on a working model risk function; without it, the strategy is a set of intentions without a control framework.

9.2.2 AI policy and ethics

A defined function, typically reporting into the chief risk officer or the chief data officer, owns the firm's AI policy, the standards models must meet before deployment, the assurance framework, and the ethical review of AI use cases. In smaller firms this may be a single role; in larger firms it is a function. Its independence from the first line is important: it should be able to block deployments that do not meet the standards.

9.2.3 Conduct and consumer-duty oversight

Where the firm operates under a conduct regime, an explicit role connects HITL design to conduct outcomes. This function reviews the customer-facing AI use cases, monitors fair-value, vulnerable-customer and complaints metrics, and is the link between the firm's HITL strategy and its conduct accountability.

9.3 Governance bodies

Three governance bodies are typically required.

An AI risk committee, typically chaired at chief risk officer or chief data officer level and reporting to the executive committee, sets policy, approves material AI deployments, and reviews aggregate AI risk on a defined cadence. It is the point at which AI risk integrates with the firm's broader risk appetite.

An AI ethics or design review board reviews proposed AI use cases, particularly those affecting customer outcomes, against the firm's stated principles. It is typically smaller than the risk committee and includes external voices, and its role is preventative rather than monitoring.

An incident response capability handles AI failures: model accuracy regressions, fairness incidents, customer harms, and any incidents requiring regulatory notification. It is connected to the firm's broader operational resilience and incident management arrangements, not stood up in isolation.

9.4 The data and technology layer

The HITL strategy depends on data and technology capabilities that are easy to underestimate. These include a model inventory that records every model in use and its associated metadata; a monitoring platform that runs the accuracy, fairness and drift metrics on a defined cadence; the interaction surfaces through which front-line staff work with AI outputs; the data infrastructure that captures human interventions as labelled data; the explainability tooling that supports contestability; and the audit infrastructure that demonstrates the operation of the strategy to internal and external reviewers.

These capabilities are mostly built on procured platforms, configured to the firm's specific requirements. The build-versus-buy decision is typically less consequential than the design and configuration decisions taken once the platform is in place.

10. Regulatory expectations

HITL design in insurance does not happen in a regulatory vacuum. Several regimes bear directly on the design choices the firm makes.

10.1 The EU AI Act

The European Union's AI Act classifies insurance-relevant AI systems by risk and imposes obligations accordingly. Systems used for risk assessment and pricing in life and health insurance are classified as high-risk, with obligations around risk management, data governance, transparency, human oversight, accuracy, robustness and cybersecurity. These obligations map closely to the components of a credible HITL strategy as set out in Part I; a firm whose strategy is well-designed will be substantially on the way to compliance.

The AI Act's human oversight requirement is particularly aligned with HITL: deployers of high-risk systems must ensure that humans can oversee the system effectively, intervene where necessary and disregard or reverse the system's output. The strategic question for an insurer operating in Europe is therefore not whether to design human oversight but how, and the framework set out in Part I is one defensible answer.

10.2 The FCA Consumer Duty

In the United Kingdom, the Consumer Duty creates obligations on firms to deliver good outcomes for retail customers, to support customers in pursuing their objectives, and to act in good faith. The Duty's outcomes framework, covering products and services, price and value, consumer understanding, and consumer support, applies directly to AI-influenced customer journeys. A HITL strategy designed to deliver contestable, explicable, fair-value decisions is the operating model in which Consumer Duty compliance lives.

The Duty is not prescriptive about how AI is used, but it does require firms to be able to demonstrate that they are delivering the four outcomes. The data infrastructure that captures human interventions and customer outcomes in a HITL operating model is therefore also the data infrastructure that evidences Consumer Duty compliance.

10.3 Solvency II and the prudential regimes

Solvency II Pillar 2, and equivalent prudential regimes in other jurisdictions, impose model risk management requirements on capital, reserving and pricing models. The introduction of AI into these areas extends the scope of the existing model risk management framework rather than requiring a new one. The HITL strategy in these areas is largely a matter of integration: ensuring that the firm's existing

model risk function expands its scope appropriately and that the governance bodies receive consolidated AI risk reporting.

10.4 GDPR and data protection

Article 22 of the GDPR, and equivalent provisions in other jurisdictions, gives data subjects the right not to be subject to decisions based solely on automated processing that produce legal or similarly significant effects on them, subject to exceptions. Insurance decisions, particularly underwriting and claims decisions, generally fall within the scope. The practical effect is that for any AI-decided customer-facing decision of significant effect, the firm must either rely on an exception, must obtain explicit consent, or must provide for meaningful human review. A HITL strategy that includes human-as-fallback for adverse decisions and human-in-command for the most material ones is the most defensible response to Article 22.

10.5 State-level insurance regulation in the United States

In the United States, insurance regulation is state-led and the National Association of Insurance Commissioners' model bulletins on AI and predictive models are influential. The NAIC framework emphasises governance, risk management, testing for unfair discrimination and consumer protection. The HITL strategy maps to this framework along similar lines to its mapping to European regimes: the components of governance, monitoring, fairness testing and human oversight are common to both.

10.6 The convergence

Across these regimes there is substantial convergence on a small number of expectations. AI systems affecting consumers must have meaningful human oversight. Decisions must be explainable. Adverse decisions must be contestable. Fairness must be monitored. Material AI use must be governed. A credible HITL strategy, designed against the framework in Part I, will be substantially compliant with all of them. A firm that has not designed such a strategy will find itself building one in response to whichever regime audits it first, which is the more expensive way to do it.

> *Regulators are converging on the question that the framework asks: for each decision, who is the accountable human, what evidence informed it, and how can the affected party contest it?*

11. An implementation roadmap

This section sets out an indicative roadmap for an insurer implementing the strategy from a near-greenfield starting point. The timeline is illustrative; firms with mature model risk management and developed AI capabilities will move faster, and firms in earlier stages may need longer.

11.1 The first ninety days: orientation and inventory

The first phase is establishing the position. Inventory the AI systems in use across the firm. Inventory the customer-facing decisions in scope. Map each AI system to the decisions it informs and the pattern under which it currently operates. Identify the gaps between current practice and the components of a credible strategy.

Outputs include the AI and decision inventory, an as-is map of patterns, and a prioritised gap list. The exercise is sometimes treated as paperwork; it is in fact the foundation of the strategy and is the single most useful thing a firm can do in its first ninety days.

11.2 Months three to nine: design

The second phase is design. For each priority decision class, define the indicated pattern, the triage design, the interaction surface and the role configuration. Define the monitoring framework, including the metrics that will be collected and the cadence at which they will be reviewed. Stand up the AI risk committee and the design review board if they do not already exist. Begin retraining and redeploying front-line roles for the new operating model.

Outputs include the HITL design for each priority decision class, the monitoring framework, the governance arrangements, and an implementation plan for the affected systems and roles.

11.3 Months six to eighteen: build and bed in

The third phase, which overlaps with the second, is implementation. Build or configure the interaction surfaces and the monitoring platform. Deploy the redesigned roles. Operationalise the governance bodies. Run the new operating model alongside the existing one for the highest-risk decision classes before cutting over.

Expect the bed-in period to be longer than the planning indicates. HITL operating models depend on tacit knowledge that accumulates only through operation: which cases the AI handles well, which it handles poorly, what the front-line staff need from the interaction surface, what the customer experience actually feels like. Build in a period of explicit learning, with metric-driven review and a willingness to adjust the configuration.

11.4 Months twelve to twenty-four: maturity

The fourth phase is maturity. By the end of the second year the firm should have a working HITL operating model across the priority decision classes, an established assurance regime, and the data infrastructure to demonstrate the operation of the strategy to internal stakeholders, regulators and customers. The focus shifts from establishing the framework to extending it to lower-priority decision classes and improving its performance on the established ones.

11.5 Anti-patterns to avoid

A few specific anti-patterns recur and are worth naming.

- Treating HITL as a compliance project. The strategy fails if it is owned by the compliance function in isolation and produces documents rather than an operating model.
- Buying a platform before designing the strategy. The right tool depends on the design; the right design does not depend on the tool.
- Underinvesting in the interaction surface. The surface is the place where most of the actual HITL value is captured or lost, and it is often the most underfunded element of the build.
- Treating retraining the workforce as optional. The new operating model requires new skills, new judgement and new metrics, and not investing in them produces oversight theatre at scale.
- Centralising HITL decisions away from the people who will operate them. The strategy benefits from central coordination but its credibility depends on operational ownership at the level where the decisions are made.

12. Conclusion

This document has argued that human-in-the-loop is an enterprise strategy, not an operational pattern. It has set out a general framework consisting of canonical patterns, a decision-driven rubric for choosing among them, and the six components that a credible strategy must include. It has applied that framework to the insurance industry, walking through the principal decisions in the value chain and identifying the indicated patterns, the specific considerations and the regulatory context.

Three themes run through the analysis. The first is that HITL design follows the decision, not the model. The capability of the underlying AI is one input into the design, but it is the consequence, reversibility and contestability of the decision that should drive the pattern. The second is that HITL is a portfolio. A mature operating model uses human-in-command, human-on-the-loop, human-out-of-the-loop and human-as-fallback concurrently, matched to the decisions at hand. The third is that the assurance regime around HITL is at least as important as the design. A well-designed strategy that does not produce evidence of its own operation will not survive contact with the first material incident.

For insurance specifically, the convergence of regulatory expectation around meaningful human oversight, explainability, contestability and fairness means that the strategic question is settled in direction even if the implementation details are not. The firms that build credible HITL operating models in the next two to three years will find themselves substantially aligned with the regulatory regimes that follow. The firms that do not will be retrofitting under examination, which is the costlier path.

The framework offered here is methodological. It is intended to be reusable across decisions, lines of business and regulatory regimes. The insurance application is illustrative of how the framework lands when grounded in a specific industry. The same logic, adjusted for context, applies in banking, healthcare, public services and any other sector where AI is now sufficiently capable to do material work but not sufficiently trustworthy to do it unsupervised.

> *The destination is not full automation, and it is not retained manual operation. It is an operating model in which the AI does the work it does best, the humans do the work they do best, and the seam between them is designed deliberately rather than left to chance.*

About this document

It is methodological in intent and is not legal, regulatory, actuarial or financial advice. Specific firms applying the framework should adapt it to their own circumstances, risk appetite, regulatory environment and operating model, and should take appropriate professional advice on points of legal

and regulatory interpretation. The insurance application in Part II is intended to illustrate the framework in a specific industry context and is not a prescription for any particular firm.